Area and differential equations (3)

Q1. A function $y = f(x)$ satisfies the differential equation $f(x)\sin 2x - \cos x + (1+\sin 2x)f'(x) = 0$ with initial condition $y(0) = 0$ then the value of $(10\, f(\frac{\pi}{2}))$ is equal to

Q2. If the area bounded by the curve $(y - \text{arc}\sin x)^2 = x - x^2$ is A, then the value of $[4A]$ is equal to ([.] is GIF)

Q3. Let $f(x) = x^2 + 6x + 1$ and S denotes the set of points (x, y) in the coordinate plane such that $f(x) + f(y) \le 0$ and $f(x) - f(y) \le 0$. The area of S is $k\pi$ then $k = ___$.

Q4. If the area bounded by $y = [\sin^2\frac{x}{2} + \sqrt{\cos\frac{x}{2}}]$ and $y = |\sin x - \frac{1}{2}|$, $x = 0$, $x = \pi$ is S then $[2S] = __$ (where [.] is GIF).

Q5. The area of the region bounded by the curves $y = \frac{4-x^2}{4+x^2}$, $25y^2 = 9x$ and $y = \frac{3}{5}|x| - \frac{6}{5}$ which contains $(1, 0)$ point in its interior is $\pi - a\tan^{-1}\frac{1}{2} + \frac{1}{b}$ where $b - 2a$ is equal to (a, b are positive integers)

Q6. The minimum area bounded by the function $y = f(x)$ and $y = \alpha x + 9$ ($\alpha \in R$) where f satisfies the relation $f(x+y) = f(x) + f(y) + y\sqrt{f(x)}$ $\forall x, y \in R$ and $f'(0) = 0$ is $9A$, value of A is

Q7. If the area bounded by the curves $y = -x^2 + 6x - 5$, $y = -x^2 + 9x - 3$ and the line $y = 3x - 15$ is $\frac{73}{d}$, where $x > 1$ then the value of d is

Q8. If the area bounded by the region $|y| + 1 = 2e^{-|x|}$ is $k(1 - \log_e 2)$ then the numerical value of k is

Q9. If the area enclosed between the curves $y = \sin^4 x \cos^3 x$, $y = \sin^2 x \cos^3 x$ between $x = 0$ and $x = \frac{\pi}{2}$ is A, then which of the following is/are incorrect?

(A) $A = \int_0^1 t^2(1-t^2)\,dt$ (B) $A = \int_0^1 t^2(1-t^2)^2\,dt$

(C) $A = \int_0^1 t(1-t^2)^2\,dt$ (D) $A = \frac{\pi}{2}\int_0^1 t(1-t^2)\,dt$

Q10. Solution of the differential equation

$$\frac{x + y\frac{dy}{dx}}{y - x\frac{dy}{dx}} = \frac{x\sin^2(x^2+y^2)}{y^3}$$ is

(A) $-\cot(x^2+y^2) = \left(\frac{x}{y}\right)^2 + c$ (B) $\frac{y^2}{x^2+y^2c} = -\tan^2(x^2+y^2)$

(C) $-\cot(x^2+y^2) = \left(\frac{y}{x}\right)^2 + c$ (D) $\frac{y^2+x^2c}{x^2} = -\tan^2(x^2+y^2)$

Q11. If the area enclosed by $y^2 = 4ax$ and line $y = ax$ is $\frac{1}{3}$ sq. units, then the roots of the equation $x^2 + 2x = a$ are

A) -4 (B) 2 (C) -2 D) 8.

Q12. If $A\left(\frac{3}{\sqrt{2}}, \sqrt{2}\right)$, $B\left(-\frac{3}{\sqrt{2}}, \sqrt{2}\right)$, $C\left(-\frac{3}{\sqrt{2}}, -\sqrt{2}\right)$ and $D(3\cos\alpha, 2\sin\alpha)$ are four points, then the value of α for which the area of quadrilateral ABCD is maximum, $\left(\frac{2\pi}{2} \le \alpha \le 2\pi\right)$ is (in sq. units) is K, where $[K]$ is greater than or equal to

A) 2 B) 3 C) 4 (D) 5.

Q13. Let $y(x)$ be a solution of the differential equation $(1+e^{-x})\frac{dy}{dx} - ye^{-x} = 2$. If $y(0) = 3$, then which of the following is/are true?

A) $y(-2) = 0$ (B) $y(-3) = 0$ (C) $y(x)$ has a critical point in the interval $(-5, -4)$

(D) $y(x)$ has no critical point in the interval $(-5, -4)$.

Q14. Let $g(x) = \frac{192x^3}{2+\sin^4 \pi x}$ for all $x \in R$ with $g\left(\frac{1}{2}\right) = 0$. If A be the area of the region bounded by $y = g(x)$, with x-axis and between the ordinates $x = \frac{1}{2}$ and $x = 1$ such that $\alpha \le A \le \beta$, then the possible values of $\alpha \& \beta$ are

(A) $\alpha = 14, \beta = 26$ (B) $\alpha = \frac{1}{12}, \beta = \frac{1}{2}$ (C) $\alpha = -12, \beta = 0$ (D) $\alpha = 1, \beta = 12$

Q15. Consider the hyperbola $A: x^2 - y^2 = 1$ and a circle B with centre $N(a_2, 0)$. Suppose that A and B touch each other at a point $P(\alpha_1, \beta_1)$ with $\alpha_1 > 1$ and $\beta_1 > 0$. The common tangent to A and B at P intersects the x-axis at the point M. If (a, b) is the centroid of the triangle PMN, then which of the following is/are correct?

(A) $\frac{da}{d\alpha_1} = 1 - \frac{1}{3\alpha_1^2}$ for $\alpha_1 > 1$ (B) $\frac{da}{d\alpha_1} = 1 + \frac{1}{3\alpha_1^2}$ for $\alpha_1 > 1$

(C) $\frac{db}{d\alpha_1} = \frac{\alpha_1}{3\sqrt{\alpha_1^2 - 1}}$ for $\alpha_1 > 1$ (D) $\frac{db}{d\beta_1} = \frac{1}{3}$ for $\beta_1 > 0$

Q16. If $f(x) = e^x(\sin^6 ax + \cos^4 ax)$ where $a \in R^+$ and S_1 be the area of the region bounded by $y = f(x)$, with x-axis and between the ordinates $x = 0$ and $x = 4\pi$, S_2 be the area of the region bounded by $y = f(x)$ with x-axis and between the ordinates $x = 0$ & $x = \pi$ such that $\frac{S_1}{S_2} = A$, then which of the following is/are correct?

(A) for $a = 3$, $A = \frac{e^{4\pi} - 1}{e^{\pi} - 1}$ (B) for $a = 6$, $A = \frac{e^{4\pi} - 1}{e^{\pi} - 1}$

(C) for $a = 2$, $A = \frac{e^{4\pi} - 1}{e^{\pi} - 1}$ (D) for $a = 4$, $A = \frac{e^{4\pi} - 1}{e^{\pi} - 1}$

Q17. Let $y = f(x)$ be the continuous function in $[0, \infty)$ satisfy $\int_0^x f(x-t)\,dt = x f(x) + x^2 \int_0^1 f'(tx)\,dt \; \forall x > 0$ & $f(0) = 3$. Then which of following is/are correct

(A) Area bounded by $y=f(x)$ and x-axis in between $x=0$ & $x=3$ is $\frac{9}{2}$ (B) Area bounded by $y=f(x)$ & x-axis in between $x=0$ & $x=3$ is 9.

(C) Number of points of intersection of $y=|f(x)|$ and $x+y=4$ be 1 (D) Number of points of intersection of $y=|f(x)|$ and $y-x=4$ be 1.

Q18. The differential equation of family of

Column I	Column II
(A) circles passing through the origin and having their centres on the x-axis	(P) $xy\frac{d^2y}{dx^2}+x\left(\frac{dy}{dx}\right)^2-y\frac{dy}{dx}=0$
(B) parabolas with foci at the origin and axis along the x-axis	(Q) $y^2=x^2+2xy\frac{dy}{dx}$
(C) parabolas each of which has a latus-rectum $4a$ and whose axis are parallel to x-axis	(R) $y\left(\frac{dy}{dx}\right)^2+2x\frac{dy}{dx}-y=0$
(D) ellipse centered at the origin with an axis along x-axis	(S) $2x\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^3=a$

Q19. Let $y=f(x)$ be a curve passing through (e,e^e) which satisfy the differential equation $(2xy+xy\log_e x)dx-x\log_e x\,dy=0$, $x>0$, $y>0$.

If $g(x)=\lim_{n\to\infty} f(x)$, then $\int_{1/e}^{e} g(x)\,dx=$

Q20. Let C be the curve passing through the point $(1,1)$ has the property that the perpendicular distance of the origin from the normal at any point P of the curve is equal to the distance of P from the x-axis. If the area bounded by the curve C and x-axis in the first quadrant is $\frac{k\pi}{2}$, then the value of k is equal to

Q 21. If A is the area between the curve $y = x^6(1-x)^7$ and x-axis then the number of prime positive integral factors of A is ___

Q 22. Let the straight line $y = k$ divide the area enclosed by $x = (1-y)^2$, $x = 0$ and $y = 0$ into two parts of areas, $A_1 (0 \le y \le k)$ and $A_2 (k \le y \le 1)$ such that $\frac{A_1}{A_2} = 7$. Then $\frac{1}{k}$ is equal to

Q 23. Let the area of the region bounded by the curves $y^3 = x^2$ and $y = 2 - x^2$ be $\frac{30+\alpha}{15}$, then α equals

Q 24. Let $f(x) = \int_x^{x^2+\frac{\pi}{3}} 4\sin^2 t \, dt$ for all $x \in \mathbb{R}$ and $g : [0, \frac{1}{2}] \to [0, \infty)$ be a continuous function. For $\alpha \in [0, \frac{1}{2}]$, if $f'(\alpha) + 2$ is the area of the region bounded by $x = 0$, $y = 0$, $y = g(x)$ and $x = \alpha$, then the value of $g(0)$ is equal to

Q 25. If $y = f(x)$ be a curve passing through $(1, 3)$ satisfying the differential equation $(x^2 + xy + 4x + 2y + 4)\frac{dy}{dx} = y^2 (x > 0)$, then the number of solutions of the equation $f(x) = x + 2$ is ___

Q 26. If A be the area of the region represented by $\{(x, y) \in \mathbb{R}^2 : y \ge \sqrt{|x+3|}, 5y \le x + 9 \le 15\}$, then the value of 4A is equal to

Q27. If a curve $y = a\sqrt{x} + bx$ passes through the point (1,2) and the area bounded by the curve, the line $x = 4$ and x-axis is 8 square units then which of the following is/are true?

(A) $a = 3$ (B) $a = -3$ (C) $b = 1$ (D) $b = -1$.

Q28. Area bounded by the curve $y = \log_e x$, $y = 0$ & $x = 3$ is

(A) $(\log_e 9 - 2)$ sq. units (B) $(\log_e 27 + 2)$ sq. units

(C) $\log_e\left(\frac{27}{e^2}\right)$ sq. units (D) greater than 1 sq. unit

Q29. If $\left(\frac{x+y-1}{x+y-2}\right)\frac{dy}{dx} = \left(\frac{x+y+1}{x+y+2}\right)$, given that $y = 1$ when $x = 1$, then which of the following is/are not a solution of the given differential equation?

(A) $\ln\left|\frac{(x-y)^2-2}{2}\right| = 2(x+y)$ (B) $\ln\left|\frac{(x+y)^2-2}{2}\right| = 2(x-y)$

(C) $\ln\left|\frac{(x-y)^2+2}{2}\right| = 2(x+y)$ (D) $\ln\left|\frac{(x+y)^2+2}{2}\right| = 2(x-y)$.

Q30. Two lines drawn from the point P(4,0) divide the area bounded by the curve $y = \sqrt{2}\sin\frac{\pi x}{4}$ and x-axis between the lines $x = 2$ and $x = 4$ into three equal parts.

(A) Sum of the slopes of the drawn lines is equal to $-2\sqrt{2}/\pi$ (B) Sum of the slopes of the drawn lines is equal to $-\frac{\sqrt{2}}{\pi}$

(C) Product of the slopes of the drawn lines is equal to $\frac{16}{9\pi^2}$. (D) Product of the slopes of the drawn lines is equal to $\frac{8}{9\pi^2}$.

Q31. The area bounded between the curves $y^2 = 4x$, $x^2 + y^2 = 8x$ and $y \geq 0$ is less than equal to

(A) $\frac{32}{3} + 4\pi$ (B) $\frac{41}{3} - 4\pi$ (C) $4\pi - \frac{32}{3}$ (D) $4\pi - \frac{2}{3}$.

Q32. The area bounded by $y = \frac{\sin x}{x}$, x-axis and the ordinates $x = 0$, $x = \frac{\pi}{4}$ is

(A) ≤ 1 (B) $< \frac{\pi}{4}$ (C) $\leq \frac{\pi}{2}$ (D) $\leq \frac{\pi}{3}$.

Q33. Let $f(x) = \frac{x}{1+x^2}$ and $g(x) = \frac{1}{x(1+x^2)}$. If A_1 be the area of region bounded between $y = f(x)$, x-axis, $x = \frac{1}{e}$ and $x = \pi$, and A_2 is the area of region bounded between $y = g(x)$ x-axis and between $x = \frac{1}{\pi}$ and $x = \frac{1}{e}$. Then $|A_1 - A_2|$ is greater than or equal to

(A) 2 (B) 1 (C) $\ln \frac{3}{2}$ (D) $\ln \frac{13}{11}$.

P: Let $f(x) = \sin 4\pi x$ and $g(x) = \ln x$. Let $x_0, x_1, x_2, \ldots, x_{n+1}$ be the roots of $f(x) = g(x)$ in increasing order. Then answer the following

Q34. The value of n is less than or equal to

(A) 2 (B) 3 (C) 4 (D) 5

Q35. The whole area (approximately) bounded by $y = f(x)$, $y = g(x)$ and $x = 0$ is less than or equal to

(A) $\frac{17}{8}$. (B) $\frac{15}{8}$ (C) $\frac{13}{8}$ (D) $\frac{11}{8}$.

P: Let us consider three functions $f(x)$, $g(x)$, $h(x)$ defined as

$f(x) = \sin^{-1}|\sin x|$, $x \in [0, 2\pi]$, $g(x) = \cos^{-1}|\cos x|$ $x \in [0, 2\pi]$

$$h(x) = \begin{cases} x, & x \in [0, \frac{\pi}{2}] \\ x - \frac{\pi}{2}, & x \in (\frac{\pi}{2}, \pi] \\ x - \pi, & x \in (\pi, \frac{3\pi}{2}] \\ x - \frac{3\pi}{2}, & x \in (\frac{3\pi}{2}, 2\pi] \end{cases}$$

Q36. Number of integral solution of $f(x) = h(x)$ is greater than or equal to

(A) 1 (B) 2 (C) 4 (D) ∞

Q37. Area bounded by the curves $f(x)$ and $g(x)$ is less than or equal to

(A) $\frac{\pi^2}{4}$ (B) $\frac{3\pi^2}{4}$ (C) $\frac{9\pi^2}{4}$ (D) $\frac{5\pi^2}{4}$.

Q38. The order, degree of the differential equation satisfying $\sqrt{1+x^2} + \sqrt{1+y^2} = a(x\sqrt{1+y^2} - y\sqrt{1+x^2})$ is

1) 1, 1 2) 2, 1 3) 3, 2 4) 0, 1

Q39. The solution of

$y(xy+1)\,dx + x(1+xy+x^2y^2)\,dy = 0$ is

1) $2x^2y^2 \log y - 2xy - 1 = cx^2y^2$ 2) $2x^2y^2 \log y + 2xy - 1 = cx^2y^2$ 3) $2xy \log x + 2xy + 1 = cx^2y^2$ 4) $2xy \log y - 2xy - 1 = cx^2y^2$.

Q40. At present a firm is manufacturing 2000 items, it is estimated that the rate of change of production P w.r.t additional number of workers x is given by $\frac{dp}{dx} = 100 - 12\sqrt{x}$. If the firm employs 25 more workers, then new level of production of items is

1) 2500 2) 3000 3) 3500 4) 4500.

Q41. The area bounded by the two branches of curve $(y-x)^2 = x^3$ and the straight line $x = 1$ is

1) $\frac{1}{5}$ 2) $\frac{3}{5}$ 3) $\frac{4}{5}$ (4) $\frac{8}{4}$.

Q42. Let I be the purchase value of an equipment and V(t) be the value after it has been used for t years. The value V(t) depreciates at a rate given by differential equation $\frac{dV(t)}{dt} = -k(T-t)$

where $k > 0$ is a constant and T is the total life in years of the equipment. Then the scrap value V(T) of the equipment is

1) $I - \frac{k(T-t)^2}{2}$ 2) e^{-kT} (3) $T^2 - \frac{1}{k}$ 4) $\frac{I - kT^2}{2}$

Q43. P is a variable point in the square formed by the lines $x = \pm 1$ and $y = \pm 1$. P moves such that distance from the origin is less than its distance from any side of the square. The area traced by the points P is

1) $\frac{4}{3}(4\sqrt{2}+1)$ 2) $\frac{4}{3}(4\sqrt{2}-1)$ 3) $\frac{4}{3}(4\sqrt{2}-3)$ 4) $\frac{4}{3}(4\sqrt{5}-5)$

Q44. The area bounded by the curves

$|y+x| \le 1$, $|y-x| \le 1$ and $2x^2 + 2y^2 \ge 1$ is

1) $(2+\frac{\pi}{2})$ 2) $(2-\frac{\pi}{2})$ 3) $(4-\frac{\pi}{2})$ 4) $(4+\frac{\pi}{2})$.

Q45. Let f be a differentiable function $f'(x) = f^2(x)$ and $f(0) = -1$. If $\frac{\sqrt{2}}{\pi}\int_0^1 \left(\frac{(1-x)}{f(x)(x^4+1)} + \frac{2}{x^4+1}\right)dx = \frac{p}{q}$ where $p, q \in N$ then find least value of $(p+q)$.

Q46. Let k be a positive integer and f(x) be a polynomial with integer coefficients satisfying $2\int_1^x f(t)dt + x^k = x f(x)$, where $x \ge 1$. Find the sum of all possible values of k.

Q47. If $f: R^+ \to R$ be a differentiable function satisfying $f(x) = x\ln x + x - \int_1^x \frac{f(t)}{t+\ln t}\,dt$ $\forall x \in R^+$, then find the value of $f'(e)$ + $f''\left(\frac{1}{e}\right)$?

Q48. Let f be a differentiable function on R and satisfying $f(x) = -(x^2 - x + 1)e^x + \int_0^x e^{x-y} \cdot f'(y)\,dy$. If $f(1) + f'(1) + f''(1) = ke$, where $k \in N$, then find k.

Q49. Let $f: R^+ \to R$ be a differentiable function satisfying $f(x) = e + (1-x)\ln\left(\frac{x}{e}\right) + \int_1^x f(t)\,dt$ $\forall x \in R^+$. If the area enclosed by the curve $g(x) = x(f(x) - ex)$ lying in the fourth quadrant is A, then find the value of A^{-1}.

Q50. If a function f satisfies the relation $f(x)f''(x) - f(x)f'(x) = (f'(x))^2$ $\forall x \in R$ and $f(0) = f'(0) = 1$, then identify which of the following statement(s) is (are) correct?

(A) Equation $f(x) = e^x$ has two solutions

(B) Equation $f(x) = e^x$ has exactly one solution

(C) $\lim_{x \to -\infty} f(x) = \frac{1}{e}$

(D) $\lim_{x \to 0} \frac{f(x) - 1}{x} = 1$.

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Area and differential equations (3)

1. 4
2. 3
3. 8
4. 4
5. 2
6. 8
7. 6
8. 4
9. A, C, D
10. A, B
11. A, B
12. A, B, C, D
13. B, C
14. D
15. A, C, D
16. A, B, C, D
17. B, C
18. A-Q, B-R, C-S, D-P
19. 0
20. 1
21. 5
22. 2
23. 2
24. 6
25. 1
26. 6
27. A, D
28. C, D
29. A, C, D
30. A, C
31. A, C, D
32. A, B, C, D
33. ~~A, B, C, D~~ B, C, D
34. ~~B, C, D~~ A, B, C, D
35. A, B, C, D
36. A, B
37. A, B, C, D